LotBoNC: Novel Botnet Traffic Classification under Long-tailed Distributions

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Abstract

Botnets are a persistent cyber threat, leveraging global infrastructures to launch large-scale attacks. Yet, most existing classification methods are evaluated under balanced and closed-set assumptions, which fail to capture real-world conditions. In practice, botnet traffic is both long-tailed and open-world: unknown variants continually emerge, and rare threats are buried under dominant traffic, often evading detection. To reflect real-world conditions, we define a deployment-oriented setting where unlabeled traffic follows a long-tailed distribution, with dominant known classes in the head and rare novel botnet variants in the tail. We propose **LotBoNC**, a unified framework for encrypted traffic classification under long-tailed open-world conditions. LotBoNC first performs self-supervised pre-training to learn transferable representations, then applies entropy-regularized optimal transport to assign pseudo-labels aligned with estimated class priors. An EM-style loop iteratively refines prototypes and priors, improving class separation between frequent and rare categories. We evaluate LotBoNC on three public encrypted traffic datasets with diverse long-tailed scenarios. LotBoNC consistently outperforms prior state-of-the-art methods and accurately classifies known botnets and discovers unseen botnet variants in diverse, imbalanced open-world scenarios.

Keywords

Network Security, Machine Learning, Out-of-Distribution, Intrusion Detection

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1 Introduction

Botnets are pervasive and adaptive cyber threats, enabling large-scale attacks such as DDoS [44], credential stuffing, spam propagation, and malware distribution [17, 29]. These attacks are driven by globally distributed infrastructures of compromised devices, ranging from enterprise systems to consumer IoT that generate persistent, high-volume traffic [28, 67]. To evade detection, botnets rapidly evolve into polymorphic variants and adopt encrypted communications [19, 20, 56], rendering signature-based methods and deep packet inspection increasingly ineffective [1, 37]. As a result, accurate traffic classification has become a critical requirement for modern cyber defense and threat intelligence [41, 47, 65].

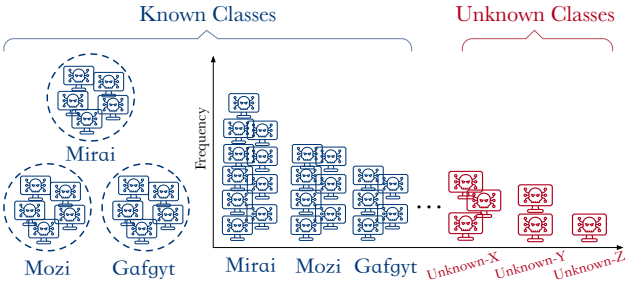


Figure 1: Illustration of long-tailed botnet traffic in the open-world setting. A few known botnet families (e.g., Mirai, Mozi and Gafgyt) have abundant labeled samples, while many emerging unknown botnets appear sparsely and without labels. The distribution is highly imbalanced, and the goal is to accurately discover these unknown low-frequency classes within unlabeled traffic.

In real-world deployments, botnet traffic exhibits two compounding characteristics: *a continuous emergence of unseen botnet variants and a highly imbalanced distribution of class frequencies.* Network streams often contain a mix of benign flows, known botnets, and rare unknown/novel families [3], forming an open-world setting with long-tailed distribution [19, 20]. In this structure, as shown in Figure 1, a few dominant botnet classes (e.g., Mirai, Mozi, Gafgyt) account for most labeled traffic, while rare and emerging threats appear sparsely and without annotation information, often buried deep in the tail. This imbalance introduces two coupled challenges: *(i) unknown classes are difficult to detect without supervision, and (ii) long-tailed distributions bias models toward head classes, causing rare unknown botnets to be misclassified as known malware or even benign traffic.* These realistic settings mirror operational environments, where rare threats are overwhelmed by dominant traffic and frequently go undetected.

Recent research on encrypted traffic classification has explored different paradigms [37, 61, 72, 74]. However, these methods assume that all classes are known in advance and labels are sufficiently represented during training, evaluating models in closed-world or balanced-class settings. Although these assumptions simplify

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evaluation, they fail to capture the realistic nature of real-world deployments, where new botnet families emerge infrequently and annotation supervision is sparse or missing. Some recent works explore novel class discovery on rare traffic [64], but fail to consider class imbalance, which is critical for modeling realistic network conditions. Consequently, the ability of existing methods to generalize under realistic, long-tailed distribution remains limited.

To address the gap, we propose **LotBoNC**, short for **Lo**ng-**t**ail **Bo**tnet **N**ovel **C**lassification, a unified framework for encrypted traffic classification that jointly tackles the challenges of novel class discovery and severe class imbalance. LotBoNC is designed for scenarios with both labeled and unlabeled traffic, where labeled samples come from known classes and unlabeled data may contain novel botnet variants. **LotBoNC** begins with self-supervised pre-training to learn transferable representations from raw traffic sequences. It then performs pseudo-label assignment using entropy-regularized optimal transport [11, 48], aligning predictions with estimated class priors. A central component of LotBoNC is an EM-style iterative refinement loop. In each round, pseudo-labels for unlabeled data are assigned via optimal transport, aligned with estimated class priors. These assignments are then used to update class prototypes and priors, progressively improving class separation, especially between dominant and rare classes in open-world and long-tailed distribution. Source code: https://github.com/watsonhu-hpi/LotBoNC

Our main contributions are summarized as follows:

- We introduce a realistic setting for encrypted traffic analysis, where unlabeled data exhibits a long-tailed distribution and unknown botnet classes are rare, making the setting inherently imbalanced and challenging.
- We propose **LotBoNC**, a unified framework that simultaneously classifies known botnet traffic and discovers unknown classes in long-tailed distribution. It combines self-supervised pre-training, optimal transport pseudo-labeling, and iterative refinement to enable rare-class discovery in open-world, long-tailed traffic.
- We demonstrate that LotBoNC consistently outperforms existing methods across three public botnet traffic datasets, achieving state-of-the-art results in both labelled and unlabelled data classes, up to 19.4% in ISCX-2014, 17.2% in CTU-13 and 14.7% in IoT-23.

2 Background

This section introduces the key foundations necessary to understand our problem setting and technical approach. We first describe the characteristics and challenges of botnet traffic classification. Next, we present the novel category discovery (NCD) framework, which enables simultaneous classification of known threats and discovery of unknown attack types in unlabeled data. Finally, we explain the long-tailed learning setting, which reflects the severe imbalance between common and rare classes in real-world network traffic.

2.1 Botnet Traffic Classification and Detection

Botnets remain one of the most persistent and damaging threats to modern networks, enabling adversaries to conduct distributed denial-of-service (DDoS) attacks, deliver malware at scale, and exfiltrate sensitive information. Their infrastructure spans a wide range of compromised devices, from enterprise servers to consumer IoT endpoints, making them highly distributed and difficult to disrupt [58]. As botnets adopt more evasive and adaptive strategies, detecting their presence in real-world network traffic has become increasingly complex.

Traditional detection techniques based on rule matching or deep packet inspection are no longer sufficient. The widespread use of traffic encryption, combined with protocol mimicry and fast-flux domains, has made payload-level inspection largely ineffective. In response, recent efforts have focused on analyzing flow-level metadata, such as timing patterns, connection frequencies, and statistical features of encrypted streams. While this shift enables detection under encryption, it introduces new challenges: botnet command-and-control (C2) traffic often mimics benign services in volume, timing, and protocol usage [14].

To overcome limitations of rule-based detection, early methods in traffic classification used handcrafted statistical features with traditional classifiers such as support vector machines, decision trees, or naïve Bayes, often underperforming in noisy or encrypted environments [46]. Deep learning marked a substantial advancement: recurrent neural networks (RNNs) demonstrated superior performance in detecting botnet traffic in smart-home environments by learning hierarchical representations [49]. Hybrid CNN–LSTM models further improved detection by extracting spatial then temporal patterns from encrypted traffic flows [2]. More recently, transformer-based architectures have demonstrated better performance. [43] applies the transformer mechanism to flow-level features for intrusion detection, capturing long-term dependencies across traffic sessions. Similarly, Vision Transformers (ViT) have been adapted for IoT botnet detection, converting flow data into image-like inputs for classification tasks [66]. Additionally, ensemble deep learning frameworks combining CNN and GRU layers have enhanced detection performance in IoT botnet scenarios, especially when optimized with advanced algorithms like [50]. These approaches collectively demonstrate the increasing sophistication of machine learning in handling encrypted, adaptive, and low-visibility botnet behaviors.

2.2 Novel Category Discovery

In conventional closed-set classification, models assume that all test instances belong to a fixed set of known classes. However, this assumption fails in dynamic environments such as cybersecurity, where unknown botnet variants continuously appear and pose immediate threats. Novel Category Discovery (NCD) provides a principled solution to this challenge by considering a joint labeled-unlabeled learning setup, where the goal is to both classify known classes and discover unknown categories within the unlabeled data [34, 35]. Unlike unsupervised clustering, NCD leverages supervision from known classes to structure the embedding space. By aligning labeled and unlabeled examples in feature space, models can both preserve separation of known classes and induce

semantic grouping among unknowns. The success of NCD methods often depends on the quality of representations and the ability to propagate structural constraints into the unlabeled region. Approaches such as OpenMatch [51] and DTC have demonstrated strong performance in image domains, and prototype refinement, cluster consistency, pseudo-label sharpening which are directly applicable to network traffic analysis. In botnet classification, NCD allows simultaneous adaptation to known threat signatures and structural discovery of novel attack behaviors, even in the presence of noise and imbalance.

2.3 Long-Tailed Learning

Long-tailed label distributions frequently appear in network traffic domains, implying that a small number of categories dominate traffic while many remain rare [13, 71]. Under such skewed distributions, supervised models often overfit to head classes and fail to generalize to the tail. Addressing this requires algorithms that preserve decision boundaries for underrepresented classes without sacrificing overall performance. Popular approaches include class-balanced loss functions, margin-based calibration, and decoupled optimization of feature extractors and classifiers [10, 24, 73].

Long-tailed learning becomes significant in open-world scenarios, where data may include samples from classes that are entirely absent during training. In the context of traffic classification, these unknowns often correspond to novel botnet families or previously unseen attack behaviors. This setting imposes a dual requirement: models must learn to represent and distinguish rare known classes under severe imbalance, while also maintaining the flexibility to reject or discover unfamiliar categories without explicit supervision. Recent approaches leverage calibrated decision boundaries, uncertainty-aware scoring, and dynamic sample reweighting to mitigate the dominance of head classes and increase sensitivity to novel inputs [32, 33]. Prototype-based methods further support this objective by representing each class with an explicit centroid in feature space, which reduces the reliance on frequency-correlated classifier norms and preserves structure for tail categories [52]. Building on these ideas, frameworks such as [38, 39] integrate long-tail classification with open-set recognition and few-shot learning.

3 Related Work

Botnet detection has been extensively studied, with early efforts relying on signature matching [68], payload analysis [42], or behavioral heuristics. While effective against known threats, these systems fail to generalize to novel variants, especially as botnets evolve rapidly and increasingly use encrypted communication. Anomaly-based systems attempt to detect unknown attacks by modeling statistical deviations from normal behavior [30], but typically focus on binary detection and suffer from high false positives and poor interpretability.

To address encrypted traffic, recent works propose using flow-level statistical features as a basis for classification [40], enabling detection even when payloads are inaccessible. Supervised deep learning approaches [26, 59] have shown promise but assume all classes are known during training. This closed-world assumption is unrealistic in real deployments, where novel botnet variants are constantly emerging and most flows are unlabeled.

Open-world and unknown class detection has been studied in both security and learning communities. Security-oriented methods simulate unknown threat detection by excluding botnets during training, but often treat the task as one-class or binary detection, which fails to characterize multiple emerging threats. Recent work has explored clustering-based approaches to discover unknown attack types [21], but typically rely on offline heuristics and are not optimized for class imbalance.

Learning under class imbalance has been recognized as a key challenge in security datasets [60], where known botnets dominate and rare classes are underrepresented. Existing methods mitigate imbalance using re-weighting [10], synthetic oversampling [7], or cost-sensitive loss [55], but these assume full label availability. Semi-supervised and pseudo-labeling approaches [69] attempt to infer class structure under limited supervision, but are fragile under noisy predictions and often amplify bias toward majority classes.

Despite these advances, no prior work provides a principled framework for multiclass discovery of novel botnet traffic under encrypted, unlabeled, and long-tailed conditions. Existing approaches either ignore class granularity, assume balanced label priors, or lack mechanisms to prevent pseudo-label collapse. Our work addresses this gap by introducing a novel traffic classification framework that estimates class priors from noisy predictions and regularizes pseudo-label assignments using constrained optimal transport. This design enables robust discovery of minority botnet classes and brings practical open-world capability to encrypted traffic analysis.

4 LotBoNC Model

This section introduces our EM-based training framework for long-tailed novel botnet classification. The *E-step* assigns pseudo-labels to unlabeled traffic via entropy-regularized optimal transport, aligning them with estimated class priors to mitigate distributional bias. The *M-step* then updates class prototypes and priors through prototype-based optimization. Finally, we integrate both unsupervised and supervised objectives to promote semantic alignment across classes and preserve instance-level discriminability.

4.1 Problem Setup

We denote $(\mathbf{X_l}, Y_l)$ and $(\mathbf{X}u, Y_u)$ as samples from labeled and unlabeled probability measures $\mathbb{P}_{\mathbf{X},Y}$ and $\mathbb{Q}_{\mathbf{X},Y}$, respectively. Here, $\mathbf{X}_l \in \mathcal{X}_l$ and $\mathbf{X}_u \in \mathcal{X}_u$ are feature vectors in $\mathbb{R}^d$, and $Y_l \in \mathcal{Y}_l$, $Y_u \in \mathcal{Y}_u$ are class labels, with $\mathcal{Y}_u = \mathcal{Y}_l \cup \mathcal{Y}_n$ where $\mathcal{Y}_l$ are known classes and $\mathcal{Y}_n = \mathcal{Y}_u \setminus \mathcal{Y}_l$ are novel classes. The goal of LotBoNC is to predict labels Y_u for instances $\mathbf{X}_u$ in an unlabeled dataset $\mathcal{U}_m$ from $\mathbb{Q}_{\mathbf{X}_u,Y_u}$, by leveraging information from a balanced labeled dataset $\mathcal{L}n$ from $\mathbb{P}_{\mathbf{X}_l,Y_l}$. The unlabeled distribution $\mathbb{Q}_{Y_u}$ is imbalanced, with known classes in $\mathcal{Y}_l$ more frequent than novel ones in $\mathcal{Y}_n$ (i.e., $\mathbb{Q}_{Y_u}(i) > \mathbb{Q}_{Y_u}(j)$ for $i \in \mathcal{Y}_l$, $j \in \mathcal{Y}_n$), reflecting real-world scenarios where known traffic dominates and unknown threats are rare.

4.2 Data Preprocessing

Tokenization. To prepare raw encrypted traffic for representation learning, we adopt a burst-level tokenization strategy, following the design principles established in [12, 37, 74]. This approach captures meaningful mid-level structure within encrypted sessions and has

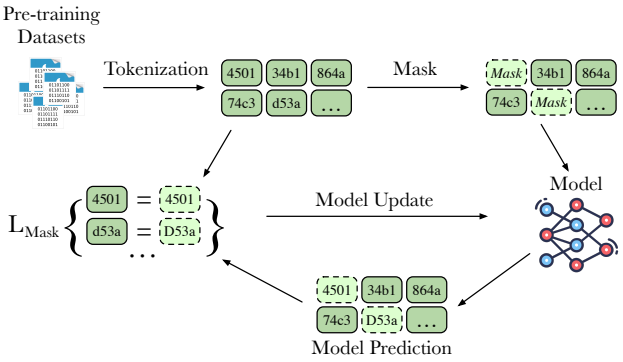


Figure 2: Overview of masked token pre-training reconstructs masked tokens from encrypted traffic sequences.

been shown effective for modeling traffic semantics. Each session is first segmented into *bursts*, which are defined as sequences of temporally contiguous packets traveling in a single direction within the same TCP or UDP flow. This directional grouping approximates application-layer boundaries and provides more structured input than raw packet sequences. Given a flow $F = \{p_1, p_2, \ldots, p_n\}$, we decompose it into bursts $B = \{b_1, b_2, \ldots, b_k\}$, where each b_i is a maximal unidirectional subsequence. Within each burst, we remove low-level protocol headers and concatenate the remaining payloads to form a clean byte stream. This stream is then tokenized into a sequence of fixed-size 2-byte *bigrams*. To support self-supervised training, we append special tokens including `[CLS]`, `[SEP]`, `[MASK]`, and `[PAD]` as needed, mirroring standard practices in masked language modeling and contrastive pretraining. The resulting tokenized burst is represented as $X = [x_1, x_2, \ldots, x_T]$, where each x_i is a 2-byte token. The input embedding for each position combines token, positional, and segment encodings. This encoding structure enables our model to leverage both spatial and contextual information within burst-level traffic segments.

Data Augmentation. We employ a principled and minimal augmentation strategy inspired by [15], introducing stochastic variability exclusively through dropout applied within the encoder. This design choice is critical in network traffic domains, where input-level perturbations can distort flow structure, particularly for tail-class samples that are both rare and structurally fragile. Concretely, given a tokenized burst-level flow X, we perform two stochastic forward passes using independently sampled dropout masks, yielding feature representations $\mathbf{h}_1 = f_{\theta_1}(X)$ and $\mathbf{h}_2 = f_{\theta_2}(X)$, where f_θ denotes the shared encoder with dropout. These views are treated as positive pairs in our contrastive objective. By introducing latent variability through encoder-level stochasticity, this approach preserves the semantic integrity of input flows while promoting robust representation learning under long-tailed and open-world settings. Importantly, it eliminates the need for heuristic, domain-specific augmentations that may compromise tail-class fidelity.

4.3 Masked and Contrastive Pre-training

Our framework incorporates two self-supervised objectives, masked modeling and contrastive learning.

4.3.1 Masked Token Pre-training. We use a masked token prediction task [12, 37] to train context-aware representations for encrypted traffic shown in Figure 2. Given a tokenized burst-level input sequence $x = [x_1, x_2, \ldots, x_T]$, we randomly select a subset of positions $\mathcal{M} \subset \{1, \ldots, T\}$ and replace the corresponding tokens with a special `[MASK]` symbol, yielding a corrupted sequence $\tilde{x}$. The model is then trained to recover the original tokens at the masked positions by minimizing the following cross-entropy loss:

$$\mathcal{L}_{\text{mask}} = -\sum_{i \in \mathcal{M}} \log p_\theta(x_i \mid \tilde{x}), \tag{1}$$

where p_θ denotes the predicted token distribution. This objective encourages the model to capture contextual dependencies within traffic flows, which is critical for handling encrypted and protocol-obfuscated data.

To enhance generalization and robustness, we pre-train this objective on a diverse collection of traffic datasets covering a wide range of encryption protocols, application types, and network behaviors. We include *CTU-MALWARE-CAPTURE* [53], *VNAT* [23], *ISCX-VPN-2016* [18], and *Ton-IoT* [45]. Dataset details are given in Section 5.1 and Table 1. This multi-source pre-training equips the model with contextually robust representations.

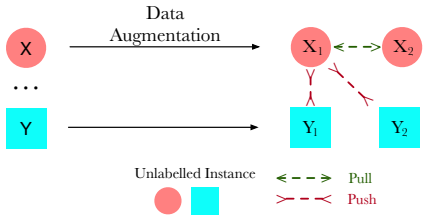


Figure 3: Overview of contrastive learning leverages unlabeled instances and augmented views to learn discriminative instance-level representations.

4.3.2 Contrastive Pre-training. To complement masked modeling, we incorporate a contrastive pre-training objective [15] aimed at learning semantically discriminative and instance-aware representations shown in Figure 3. For each tokenized burst-level input x and encoder f_θ, we obtain two latent views $h = f_{\theta_1}(x)$ and $h' = f_{\theta_2}(x)$ through stochastic forward passes. These views constitute a positive pair, while all other samples in the batch serve as implicit negatives. The model is trained using the standard InfoNCE loss [8, 57]:

$$\mathcal{L}_{\text{contrast}} = -\sum_{i=1}^{B} \log \frac{\exp(\text{sim}(h_i, h_i')/\tau)}{\sum_{j=1}^{B} \exp(\text{sim}(h_i, h_j')/\tau)}, \tag{2}$$

where $\text{sim}(u, v) = u^\top v$ denotes cosine similarity and τ is a temperature hyperparameter. This objective encourages the model to align latent representations of structurally similar flows while preserving inter-instance separation.

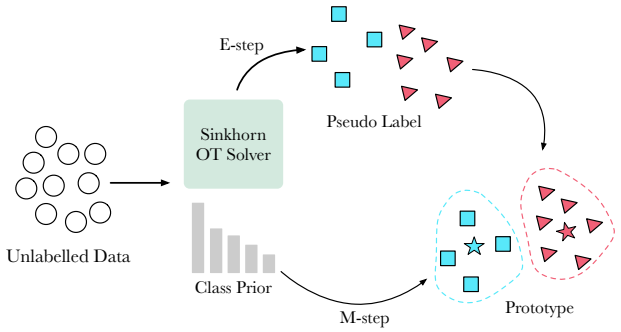


Figure 4: Overview of the EM-loop. Top: E-step assigns pseudo-labels to unlabeled traffic via entropy-regularized optimal transport aligned to estimated class priors. Bottom: M-step refines prototypes and class priors using pseudo-labels and instance embeddings to improve feature clustering and correct for class imbalance.

4.4 E-step: Pseudo-Label Assignment via Optimal Transport

In long-tailed novel botnet classification, the distribution of unlabeled traffic is inherently imbalanced, with dominant known classes and sparsely represented unknown threats. This imbalance can cause prediction collapse toward frequent classes during training. To address this, we employ entropy-regularized Optimal Transport (OT) [11, 48] for pseudo-label assignment, as illustrated in Figure 4. Specifically, OT aligns the model's predictions with an estimated class prior, enforcing balanced label assignments even under severe skew. Given a batch of m unlabeled samples, let $\mathbf{P} \in \mathbb{R}^{m\times K}$ denote the model's softmax outputs over K classes (including both known and unknown). We define a target class prior $\mathbf{r} \in \Delta^K$ and a uniform distribution over samples $\mathbf{w} \in \Delta^m$. The resulting pseudo-label assignment matrix $\mathbf{A} \in \mathbb{R}^{m\times K}$ is computed by solving the following entropy-regularized OT problem:

$$\min_{\mathbf{A}\in\Pi(\mathbf{r},\mathbf{w})} \langle \mathbf{A}, -\log \mathbf{P}\rangle - \eta H(\mathbf{A}), \tag{3}$$

where $\Pi(\mathbf{r},\mathbf{w}) = \left\{\mathbf{A} \in \mathbb{R}_+^{m\times K} \;\middle|\; \mathbf{A}^\top \mathbf{1}_m = \mathbf{r},\ \mathbf{A}\mathbf{1}_K = \mathbf{w}\right\}$ ensures marginal constraints, and $H(\mathbf{A}) = -\sum_{i,j} \mathbf{A}_{ij} \log \mathbf{A}_{ij}$ promotes smooth pseudo-label distributions. This formulation ensures proportional allocation of pseudo-labels aligned to class priors, effectively addressing class imbalance.

To efficiently solve the entropy-regularized optimal transport problem in Eq. (3), we apply the Sinkhorn-Knopp algorithm [11]. Let $\mathbf{K} = \exp(-\log \mathbf{P}/\eta)$ denote the element-wise exponentiated similarity matrix, where η is the regularization coefficient. The optimal transport matrix $\mathbf{A} \in \mathbb{R}^{m\times K}$ admits a factorized form:

$$\mathbf{A} = \mathrm{diag}(\alpha)\, \mathbf{K}\, \mathrm{diag}(\beta), \tag{4}$$

where $\alpha \in \mathbb{R}^K$ and $\beta \in \mathbb{R}^m$ are scaling vectors updated iteratively to enforce the marginal constraints. The updates follow:

$$\alpha \leftarrow \mathbf{r} \oslash (\mathbf{K}\beta), \quad \beta \leftarrow \mathbf{w} \oslash (\mathbf{K}^\top \alpha), \tag{5}$$

where $\oslash$ denotes element-wise division. This procedure is repeated until convergence. The resulting transport $\mathbf{A}$ satisfies the desired row and column marginals, and each row of $\mathbf{A}$ yields a refined pseudo-label distribution used as soft supervision in iterative training.

4.5 M-step: Prototype Alignment and Prior Estimation

The M-step refines class priors and prototype representations using the pseudo-label matrix $\mathbf{A}$ from the E-step (Figure 4). We optimize a prototype-based classification loss and apply momentum updates to both priors and prototypes.

Prototype-based classification loss. Let m be the number of unlabeled samples and K the total number of classes (including both known and novel) in the unlabeled set. Each input $\mathbf{x}_i$ is encoded into a normalized feature vector $\mathbf{v}_i = f_\theta(\mathbf{x}_i)$ via the encoder f_θ. For each class k, we maintain a prototype vector $\mathbf{c}_k \in \mathbb{R}^d$ representing the structure center of this class, and an estimated prior r_k reflecting its frequency in the unlabeled distribution.

Given the pseudo-label assignment matrix $\mathbf{A} \in \mathbb{R}^{m\times K}$ from the E-step, we define a prototype-based classification objective as:

$$\mathcal{L}_{\text{proto}} = -\sum_{i=1}^{m}\sum_{k=1}^{K} \mathbf{A}_{ik}\left[\log \frac{\exp(\mathbf{v}_i^\top \mathbf{c}_k)}{\sum_{j=1}^{K}\exp(\mathbf{v}_i^\top \mathbf{c}_j)} - \log p_k\right], \tag{6}$$

This loss encourages the embedding $\mathbf{v}_i$ to be close to the class prototype $\mathbf{c}_k$ if sample i is confidently assigned to class k (i.e., large $\mathbf{A}_{ik}$). The subtraction of $\log p_k$ acts as a prior correction term, penalizing overconfidence on frequent classes and promoting balanced assignments. This loss follows directly from an EM-based objective with prototype likelihoods and class priors. A complete derivation is provided in Appendix A.1.

Class prior estimation. Following [9, 31], we estimate class priors using the model's predicted labels over the unlabeled set. Given softmax predictions $\mathbf{P} \in \mathbb{R}^{m\times K}$, we construct an empirical distribution $\mathbf{z}$ by counting the predicted class for each sample, and apply a momentum update:

$$z_k = \frac{1}{m}\sum_{i=1}^{m} \mathbb{I}\left[k = \arg\max_j \mathbf{P}_{ij}\right], \tag{7}$$

$$r_k \leftarrow \mu r_k + (1-\mu) z_k, \tag{8}$$

where $\mu \in [0,1]$ is the momentum coefficient, and r is initialized as a uniform distribution over classes. This moving average improves the stability of class prior estimates across training epochs, particularly in the early stages when predictions are noisy.

Prototype updates. We maintain a prototype vector $\mathbf{c}_k$ for each class k using a momentum-based strategy [31]. Specifically, for each class, we aggregate embeddings from samples most confidently assigned to class k:

$$\mathbf{c}_k \leftarrow \mu \mathbf{c}_k + (1-\mu)\cdot \frac{\sum_{i=1}^{m} \mathbb{I}\left[\arg\max_j \mathbf{A}_{ij} = k\right] \mathbf{v}_i}{\sum_{i=1}^{m} \mathbb{I}\left[\arg\max_j \mathbf{A}_{ij} = k\right]}. \tag{9}$$

This ensures that prototypes evolve smoothly over time, reflecting both current embeddings and previously learned structure. We update class priors and prototypes once per epoch using the same momentum coefficient μ.

4.6 Representation Improvement

To maintain instance-level discriminability, we employ contrastive learning on unlabeled data. The augmentation strategy is mentioned in Section 4.2. Let $(\mathbf{x}_i, \mathbf{x}'_i)$ be two augmentations of the same input, with corresponding feature embeddings $(\mathbf{v}_i, \mathbf{v}'_i)$. The contrastive loss for instance i is defined as:

$$\mathcal{L}_{ins}^{(i)} = -\log \frac{\exp\left(\mathbf{v}_i \cdot \mathbf{v}'_i/\tau\right)}{\sum_m \mathbb{1}_{[n\neq i]} \exp\left(\mathbf{v}_i \cdot \mathbf{v}_m/\tau\right)}, \tag{10}$$

where $\mathbb{1}_{[n\neq i]}$ is an indicator function evaluating to 1 iff $n \neq i$, and τ is a temperature value. The overall unsupervised contrastive loss is:

$$\mathcal{L}_{\text{ins}} = \frac{1}{m} \sum_{i=1}^{m} \mathcal{L}_{\text{ins}}^{(i)}. \tag{11}$$

This loss pulls together positive pairs (same instance under augmentation) while pushing away all other instances within the batch as negatives.

In addition, we apply supervised contrastive loss [25] for the labeled data. Let y_i be the true label of sample $\mathbf{x}_i$, and define the positive set for anchor i as $\mathcal{N}(i) = \{q \neq i \mid y_q = y_i\}$. The supervised contrastive loss is given by:

$$\mathcal{L}_{sup}^{(i)} = -\frac{1}{|\mathcal{N}(i)|} \sum_{q\in\mathcal{N}(i)} \log \frac{\exp\left(\mathbf{v}_i \cdot \mathbf{v}_q/\tau\right)}{\sum_n \mathbb{1}_{[n\neq i]} \exp\left(\mathbf{v}_i \cdot \mathbf{v}_n/\tau\right)}, \tag{12}$$

where $\tau > 0$ is a temperature parameter that sharpens the similarity distribution. The loss pulls embeddings of the same class closer while pushing others apart. The denominator sums over all other labeled samples in the batch, serving as negatives.

4.7 Overall Loss Objective

The final objective integrates instance-level and class-level cues from unlabeled traffic, alongside supervised signals from labeled data:

$$\mathcal{L}_{\text{overall}} = \underbrace{\mathcal{L}_{\text{ins}} + \lambda_{\text{proto}}\mathcal{L}_{\text{proto}}}_{\mathcal{L}_{\text{unlabeled}}} + \underbrace{\lambda_{\text{sup}}\mathcal{L}_{\text{sup}}}_{\mathcal{L}_{\text{labeled}}} \tag{13}$$

The overall objective combines unsupervised and supervised components. The unsupervised loss $\mathcal{L}_{\text{unlabeled}}$ includes: (i) instance-level contrastive loss $\mathcal{L}_{\text{ins}}$ that promotes local discriminability via data augmentation and (ii) a prototype-based classification loss $\mathcal{L}_{\text{proto}}$ that aligns embeddings to class prototypes with prior correction. The supervised term $\mathcal{L}_{\text{labeled}}$ applies a contrastive loss $\mathcal{L}_{\text{sup}}$ over labeled data to enhance intra-class compactness. Hyperparameters λ_{proto} and λ_{sup} control the relative importance of each term.

5 Experiment

Our goal is to evaluate model robustness under realistic deployment conditions, where only partial supervision is available and unseen botnet variants occur with extreme class imbalance. Specifically, we simulate a network monitoring scenario in which (i) a set of known botnet families has been previously labeled through sandbox analysis or historical incident response, while (ii) new or evolving families appear unlabeled and with skewed prevalence. To reflect this asymmetry, we construct two disjoint splits for each dataset.

Algorithm 1 EM training for novel botnet traffic classification under long-tailed distributions.

Input: labeled set $\mathcal{D}_l$, unlabeled set $\mathcal{D}_u$, classifier f_θ, uniform marginals r, w, and hyperparameters λ_{proto}, λ_{sup}, μ, η
for each epoch **do**
  // sample batch $B \subset \mathcal{D}_u$ of size n_B
  for each step **do**
    // E-step: estimate pseudo-label matrix $\mathbf{A}$
    $\mathbf{P} \leftarrow \text{softmax}(f_\theta(B))$
    $\mathbf{K} \leftarrow \exp(-\log \mathbf{P}/\eta)$
    // Sinkhorn's fixed-point iteration
    for $t = 1$ to T **do**
      $\boldsymbol{\alpha} \leftarrow w \oslash (\mathbf{K}\boldsymbol{\beta})$
      $\boldsymbol{\beta} \leftarrow r \oslash (\mathbf{K}^\top \boldsymbol{\alpha})$
    end for
    $\mathbf{A} \leftarrow n_B \cdot \text{diag}(\boldsymbol{\alpha}) \cdot \mathbf{K} \cdot \text{diag}(\boldsymbol{\beta})$
    // M-step: update model parameters
    $\mathcal{L}_{\text{proto}} \leftarrow \mathcal{L}_{\text{proto}}(\mathbf{A}, \{\mathbf{c}_k\}, D_u)$ // training on D_u
    // training on D_u, $\mathbf{v}_i$, $\mathbf{v}'_i$ are augmented views
    $\mathcal{L}_{\text{ins}} \leftarrow \mathcal{L}_{\text{ins}}(\mathbf{v}_i, \mathbf{v}'_i)$
    $\mathcal{L}_{\text{sup}} \leftarrow \mathcal{L}_{\text{sup}}(\mathcal{D}_l)$ // training on D_l
    $\mathcal{L}_{\text{overall}} \leftarrow \lambda_{\text{proto}}\mathcal{L}_{\text{proto}} + \mathcal{L}_{\text{ins}} + \lambda_{\text{sup}}\mathcal{L}_{\text{sup}}$
    $\theta \leftarrow \theta - \eta\nabla_\theta \mathcal{L}_{\text{overall}}$ // update model parameters
  end for
  for each class k **do**
    $z_k \leftarrow Eq.(7)$ // calculate empirical distribution
    $r_k \leftarrow \mu r_k + (1-\mu) z_k$ // class prior update
    $\mathbf{c}_k \leftarrow Eq.(9)$ // prototype update
  end for
end for

The labelled set contains traffic from known botnet families and is explicitly class-balanced to simulate curated training corpora. The unlabelled set includes traffic from previously unseen botnet families as well as unlabeled residuals from known ones. Importantly, we impose a long-tailed distribution only on the unlabelled portion, thereby isolating the effect of imbalance in novel class discovery while keeping the supervised baseline fixed. This design reflects operational practice, where security teams have no control over the prevalence or labeling of newly emerging ones. Our experiments vary the severity of imbalance in the unlabelled set while holding the class identities fixed, allowing us to assess whether a method maintains high accuracy on known classes while correctly discovering rare variants in the tail. The remainder of this section describes the datasets, construction of long-tailed splits, baseline methods, implementation details, and evaluation metrics.

5.1 Datasets

Pre-training Datasets. We utilize four datasets from varied traffic environment to ensure comprehensive coverage. *CTU-MALWARE-CAPTURE* [53] comprises a mixture of malware, normal, and background traffic, with long-term captures categorized by malware type. *VNAT* [23] contains both encrypted and unencrypted traffic across 10 applications and 5 categories, including streaming, VoIP, chat, and file transfer. *ISCX-VPN-2016* [18] includes VPN traffic

Table 1: Summary of Datasets Used in Experiments for Pre-training and Evaluation

Pre-training		Fine-tuning	
Dataset	**# Flows**†	**Dataset**	**# Flows**†
CTU-MAL-CAP[53]	3.25×10^8	IoT-23[17]	3.25×10^6
VNAT[23]	3.17×10^6	ISCX-Bot-2014[4]	3.17×10^5
ISCX-VPN-2016[18]	2.80×10^6	CTU-13[16]	2.80×10^6
Ton-IoT[45]	2.0×10^5		

† Flow counts are reported after preprocessing steps, including filtering by size and packet count, and capping per-class samples for balance.

spanning 14 different traffic categories such as VoIP, VPN-VoIP, and P2P, and incorporates services like Skype and Facebook. Additionally, *Ton-IoT* [45] encompasses traffic ranging from DDoS attacks to scanning activities and normal operations.

Training Datasets. In training and evaluation stages, we chose three public datasets for reliable assessment of long-tailed recognition that combine realism, diversity, and public ground truth. *IoT-23* [17] was published by the Stratosphere Aposemat project in 2020. It comprises twenty malware types and three benign reference traces recorded from consumer devices. The dataset contains more than 30 GB of traffic. *ISCX-2014* [4] is from the Canadian Institute for Cybersecurity. This dataset records ten consecutive days of enterprise activity that include seven botnet families, with packet captures and 80 flow features per record, for a total of 50.0M flows. *CTU-13* [16] dataset captures thirteen distinct botnet campaigns executed in a controlled university network in 2011. The extended release encloses both PCAP and CSV representations, yielding 147.3M packets that translate to 1.3M labelled flows. The dataset includes various malware families, offering opportunities for cross domain generalisation. For each dataset, we reconstruct bidirectional flows, discard non-IP control traffic, and retain only TCP/UDP records with at least five packets. Table 1 summarizes the resulting statistics. We create two disjoint splits per dataset. We then form two disjoint splits per dataset. The labelled split consists only of known families and is class balanced by stratified sampling so that each labelled class contributes the same number of flows. The unlabeled dataset is the place where class imbalance is introduced and contains both known and unknown botnets following the long-tailed distribution. This design simulates the challenge in real-world scenario.

Long-Tailed Dataset Construction. To construct exponential long-tailed dataset, we introduce a single parameter $\gamma = n_k/n_u$, where n_k and n_u denote the number of unlabeled samples from known and unknown classes, respectively. We vary $\gamma \in \{1, 5, 10\}$ to simulate deployment scenarios with different levels of known-class dominance in the unlabeled traffic. Given C_u unknown classes, we allocate the number of samples for each class i as:

$$n_i = \max(n_{\min}, \lfloor n_{\max} \cdot \gamma^{i-1} \rfloor), \quad i = 1, \ldots, C_u,$$

where $n_{\max}$ is the sample count of the head class and $n_{\min}$ enforces a lower bound to avoid collapse in the tail. The resulting distribution is normalized to match the total n_u specified by γ.

5.2 Baselines

We compare against strong baselines used in encrypted traffic classification and recent open world recognition. All methods are trained and evaluated under the same setting: a balanced labelled split over known families and a long-tailed unlabeled split that introduces unknown botnets. Hyperparameters follow the original papers unless adaptation to mixed labelled unlabeled training is necessary. We chose the following baselines: **1DCNN** [62]. An end to end CNN for encrypted traffic classification that learns directly from raw flow derived sequences. **LSTM Att** [70]. A temporal model with attention that focuses on informative segments of encrypted flows. **TSCRNN** [36]. A hybrid that uses CNNs for spatial features and Bi LSTMs for long range temporal patterns with burst level preprocessing. **DeepTraffic** [63]. A CNN that treats traffic as images and performs malware classification through representation learning from raw traces. **ET BERT** [37]. A Transformer pre trained on large unlabeled traffic via masked BURST modeling and same origin BURST prediction, with strong few shot transfer. **NetMamba** [61]. A linear time sequence architecture tailored for traffic analysis that reduces inference latency while preserving accuracy. **RoNeTC** [64]. An open set oriented model with multi view packet representations and uncertainty estimation for unknown detection.

5.3 Experimental Setup and Hyperparameters

Pre-training Settings and Hyperparameters. During pre-training, each packet is limited to 64 bytes of data following the Ethernet layer to focus on meaningful payload information, excluding Ethernet addresses and encrypted payload content as they are irrelevant for classification tasks. The training process utilizes a batch size of 64. The Adam optimizer is employed with a learning rate of 1×10^{-5} to ensure stable and gradual learning. The model is trained for a total of 580,000 steps, with the loss function monitored to ensure convergence. A linear decay scheduling strategy with a warmup ratio of 0.1 is used to stabilize the initial training phases.

Fine-tuning Settings and Hyperparameters. We set the learning rate to 2×10^{-5} and the weight decay to 1×10^{-4} to balance optimization stability and prevent overfitting. The contrastive learning temperature was fixed at 0.5 to regulate the similarity distribution in the representation space, while a dropout rate of 0.1 was applied to reduce overfitting. The batch size was set to 32, optimizing the trade-off between computational efficiency and gradient estimation accuracy.

5.4 Evaluation Metrics

We follow the evaluation protocol of [5, 54] and compute four complementary metrics that together measure both recognition of known classes and discovery of novel ones. Let $\mathcal{D}_u$ denote the unlabeled subset, $M = |\mathcal{D}_u|$, and $\mathcal{P}(\mathcal{Y}_u)$ the set of all permutations of the unknown class indices. Given ground-truth labels y_i and model predictions $\hat{y}_i$, clustering accuracy is:

$$\text{ACC} = \max_{p \in \mathcal{P}(\mathcal{Y}_u)} \frac{1}{M} \sum_{i=1}^{M} \mathbb{1}\{ y_i = p(\hat{y}_i) \}, \tag{14}$$

where the optimal one-to-one assignment $p(\cdot)$ is obtained over all categories with the Hungarian algorithm[27].

Table 2: Performance across datasets when all classes are balanced ($\gamma = 1$). Each entry is mean ± std over multiple runs.

	Dataset	Metrics	1DCNN	LSTM-Att	TSCRNN	DeepTraffic	ET-BERT	NetMamba	RoNeTC	**LotBoNC**
$\gamma = 1$	IoT-23	Overall	48.52 ± 0.27	50.36 ± 0.48	49.45 ± 0.55	73.78 ± 0.24	75.93 ± 0.37	78.18 ± 0.31	91.34 ± 0.30	**94.48 ± 0.41**
		Labelled	82.53 ± 0.24	83.24 ± 0.30	83.05 ± 0.40	86.08 ± 0.33	92.56 ± 0.40	**98.76 ± 0.12**	97.10 ± 0.20	98.51 ± 0.31
		Unlabelled-Aware	25.69 ± 0.61	23.17 ± 0.35	22.35 ± 0.40	55.33 ± 0.13	59.99 ± 0.29	52.07 ± 0.62	86.50 ± 0.35	**90.42 ± 0.46**
		Unlabelled-Agnostic	19.43 ± 0.76	20.38 ± 0.30	19.85 ± 0.50	50.21 ± 0.15	54.72 ± 0.26	40.05 ± 0.70	91.45 ± 0.32	**88.20 ± 0.40**
	ISCX-2014	Overall	50.64 ± 0.20	49.83 ± 0.38	52.16 ± 0.10	63.77 ± 0.50	73.48 ± 0.26	65.31 ± 0.16	92.10 ± 0.28	**92.56 ± 0.33**
		Labelled	84.96 ± 0.30	84.03 ± 0.27	87.50 ± 0.60	81.21 ± 0.28	97.49 ± 0.49	**99.15 ± 0.72**	99.35 ± 0.20	99.12 ± 0.43
		Unlabelled-Aware	24.75 ± 0.57	22.16 ± 0.60	26.80 ± 0.50	37.62 ± 0.40	42.46 ± 0.36	39.85 ± 0.85	79.20 ± 0.30	**86.73 ± 0.28**
		Unlabelled-Agnostic	18.98 ± 0.31	19.55 ± 0.44	20.90 ± 0.32	33.49 ± 0.42	37.25 ± 0.34	27.06 ± 0.12	64.85 ± 0.28	**82.55 ± 0.30**
	CTU-13	Overall	52.24 ± 0.48	54.13 ± 0.30	56.95 ± 0.70	70.78 ± 0.34	76.38 ± 0.27	69.28 ± 0.18	88.02 ± 0.32	**93.29 ± 0.34**
		Labelled	82.61 ± 0.36	84.58 ± 0.50	90.27 ± 0.65	82.95 ± 0.11	95.63 ± 0.18	97.89 ± 0.46	**98.01 ± 0.22**	97.90 ± 0.25
		Unlabelled-Aware	25.84 ± 0.12	27.53 ± 0.18	31.45 ± 0.45	65.03 ± 0.48	67.52 ± 0.22	43.57 ± 0.82	80.05 ± 0.42	**88.12 ± 0.50**
		Unlabelled-Agnostic	19.79 ± 0.44	21.14 ± 0.80	27.50 ± 0.17	60.12 ± 0.50	62.46 ± 0.24	31.13 ± 0.14	71.62 ± 0.38	**80.05 ± 0.45**
$\gamma = 5$	IoT-23	Overall	57.36 ± 0.30	61.97 ± 0.55	69.93 ± 0.67	78.48 ± 0.31	83.31 ± 0.29	87.27 ± 0.04	93.93 ± 0.50	**95.56 ± 0.76**
		Labelled	91.76 ± 0.26	93.48 ± 0.32	99.43 ± 0.38	93.46 ± 0.40	97.90 ± 0.70	99.40 ± 0.13	**99.70 ± 0.15**	99.23 ± 0.08
		Unlabelled-Aware	19.55 ± 0.67	19.32 ± 0.20	22.17 ± 0.11	28.48 ± 0.34	33.62 ± 0.08	39.37 ± 0.66	51.50 ± 0.65	**75.73 ± 0.20**
		Unlabelled-Agnostic	18.36 ± 0.80	18.09 ± 0.29	17.88 ± 0.15	15.69 ± 0.68	24.33 ± 0.05	28.11 ± 0.75	38.40 ± 0.55	**73.54 ± 0.60**
	ISCX-2014	Overall	57.91 ± 0.14	52.87 ± 0.46	62.55 ± 0.08	68.60 ± 0.03	76.26 ± 0.17	72.96 ± 0.18	**93.21 ± 0.60**	93.07 ± 0.02
		Labelled	94.85 ± 0.34	92.67 ± 0.28	97.86 ± 0.61	98.53 ± 0.39	97.96 ± 0.35	**99.50 ± 0.72**	99.40 ± 0.22	98.57 ± 0.64
		Unlabelled-Aware	18.03 ± 0.61	17.26 ± 0.60	19.42 ± 0.47	23.32 ± 0.15	27.68 ± 0.31	34.18 ± 0.83	45.50 ± 0.42	**68.27 ± 0.79**
		Unlabelled-Agnostic	16.69 ± 0.29	15.85 ± 0.49	17.25 ± 0.30	18.64 ± 0.73	16.75 ± 0.06	21.53 ± 0.14	32.52 ± 0.84	**67.79 ± 0.61**
	CTU-13	Overall	58.01 ± 0.53	60.74 ± 0.29	67.37 ± 0.71	74.69 ± 0.82	80.20 ± 0.58	75.27 ± 0.20	90.56 ± 0.34	**90.55 ± 0.49**
		Labelled	91.79 ± 0.40	94.12 ± 0.55	99.29 ± 0.66	97.72 ± 0.25	98.01 ± 0.18	**98.25 ± 0.46**	99.00 ± 0.10	98.93 ± 0.38
		Unlabelled-Aware	23.90 ± 0.14	21.58 ± 0.19	25.10 ± 0.40	21.57 ± 0.11	37.48 ± 0.04	42.43 ± 0.83	47.54 ± 0.24	**72.86 ± 0.57**
		Unlabelled-Agnostic	19.24 ± 0.48	16.57 ± 0.81	21.62 ± 0.18	18.65 ± 0.09	20.82 ± 0.05	24.30 ± 0.16	34.50 ± 0.39	**71.93 ± 0.55**
$\gamma = 10$	IoT-23	Overall	59.34 ± 0.27	64.25 ± 0.52	71.07 ± 0.61	80.75 ± 0.28	86.43 ± 0.26	89.74 ± 0.03	94.66 ± 0.48	**96.93 ± 0.19**
		Labelled	84.33 ± 0.24	85.92 ± 0.31	91.24 ± 0.36	86.56 ± 0.38	96.94 ± 0.73	99.34 ± 0.12	99.65 ± 0.16	**99.93 ± 0.41**
		Unlabelled-Aware	29.06 ± 0.61	26.03 ± 0.18	31.07 ± 0.10	26.23 ± 0.31	28.42 ± 0.07	32.93 ± 0.62	45.30 ± 0.54	**60.01 ± 0.78**
		Unlabelled-Agnostic	24.84 ± 0.76	21.56 ± 0.26	21.56 ± 0.14	24.04 ± 0.65	19.24 ± 0.05	26.14 ± 0.72	37.46 ± 0.57	**56.05 ± 0.13**
	ISCX-2014	Overall	59.34 ± 0.13	54.74 ± 0.42	64.47 ± 0.07	71.16 ± 0.03	79.07 ± 0.15	75.46 ± 0.16	**94.05 ± 0.55**	93.23 ± 0.57
		Labelled	87.06 ± 0.31	84.95 ± 0.25	89.84 ± 0.62	90.33 ± 0.37	93.86 ± 0.34	**99.44 ± 0.72**	99.24 ± 0.55	98.54 ± 0.71
		Unlabelled-Aware	25.56 ± 0.59	27.06 ± 0.58	26.56 ± 0.43	24.86 ± 0.14	25.66 ± 0.30	29.27 ± 0.83	38.14 ± 0.40	**57.55 ± 0.68**
		Unlabelled-Agnostic	20.74 ± 0.28	19.46 ± 0.46	23.55 ± 0.27	21.04 ± 0.70	22.14 ± 0.05	23.35 ± 0.13	31.27 ± 0.81	**52.78 ± 0.33**
	CTU-13	Overall	61.33 ± 0.48	63.77 ± 0.26	70.34 ± 0.64	76.86 ± 0.77	81.57 ± 0.55	79.05 ± 0.18	91.66 ± 0.01	**93.93 ± 0.70**
		Labelled	84.13 ± 0.36	87.04 ± 0.52	92.86 ± 0.64	90.37 ± 0.23	95.16 ± 0.16	98.66 ± 0.46	**99.55 ± 0.09**	99.44 ± 0.08
		Unlabelled-Aware	26.04 ± 0.12	24.05 ± 0.16	23.10 ± 0.38	26.23 ± 0.10	28.16 ± 0.04	33.83 ± 0.82	42.91 ± 0.22	**60.15 ± 0.59**
		Unlabelled-Agnostic	21.03 ± 0.44	21.07 ± 0.79	19.06 ± 0.15	24.57 ± 0.08	22.24 ± 0.04	29.67 ± 0.15	34.60 ± 0.37	**56.07 ± 0.06**

First, *Overall accuracy* reflects the classification performance across both labeled and unlabeled samples. Second, *Labelled accuracy* reports the standard classification accuracy restricted to labeled data from known categories. Third, *Unlabelled-Aware* accuracy evaluates clustering performance by assuming ideal separation of novel samples and then measuring cluster-to-class alignment. Finally,*Unlabelled-Agnostic* accuracy removes this assumption and evaluates clustering on the entire unlabeled data without any indicator of whether a sample belongs to a known or unknown class. All results are averaged over five random seeds to mitigate variability caused by stochastic initialisation and data shuffling.

5.5 Comparison with SOTA

IoT-23. On the *IoT-23* dataset, LotBoNC achieves significant improvements in discovering novel classes under long-tailed distributions. In the unlabelled-aware evaluation, LotBoNC attains an accuracy of 90.42 when the class distribution is balanced ($\gamma = 1$), surpassing the strongest baseline, which reaches 86.50. As the imbalance increases, LotBoNC maintains its robustness with unlabelled-aware accuracies of 75.73 at $\gamma = 5$ and 60.01 at $\gamma = 10$, while the top baseline declines to 51.50 and 45.30, respectively. In the unlabelled-agnostic setting, LotBoNC records accuracies of 88.20, 73.54, and 56.05 for $\gamma = 1, 5$, and 10, respectively. The best baseline starts at 91.45 under balanced conditions but drops sharply to 38.40 and 37.46 as the imbalance grows. These results confirm that LotBoNC generalizes well to under-represented and unseen classes, maintaining high accuracy even under severe long-tail skew. While LotBoNC also achieves competitive overall accuracy—up to 96.93 at $\gamma = 10$—and strong labelled accuracy of 99.93 at the same setting, its most critical contribution lies in retaining reliable unlabelled predictions under increasingly imbalanced and open-world traffic distributions.

ISCX-2014. On the *ISCX-2014* dataset, LotBoNC demonstrates strong and consistent performance in discovering novel classes across varying imbalance levels. In the unlabelled-aware setting,

LotBoNC achieves an accuracy of 86.73 when the class distribution is balanced ($\gamma = 1$), outperforming the strongest baseline at 79.20. As the imbalance increases, LotBoNC maintains its advantage, reaching 68.27 at $\gamma = 5$ and 57.55 at $\gamma = 10$, while the best baseline falls to 45.50 and 38.14, respectively. In the unlabelled-agnostic metric, LotBoNC attains 82.55 at $\gamma = 1$, 67.79 at $\gamma = 5$, and 52.78 at $\gamma = 10$. The top baseline in this setting reaches 64.85 under balanced conditions but drops significantly to 32.52 and 31.27 as the imbalance increases. These results confirm that LotBoNC consistently outperforms existing methods in discovering unlabelled traffic classes.

CTU-13. The *CTU-13* results further confirm LotBoNC's effectiveness in handling long-tailed distributions for novel class discovery. In the unlabelled-aware setting, LotBoNC achieves an accuracy of 88.12 under the balanced condition ($\gamma = 1$), notably higher than the strongest baseline at 80.05. When the imbalance factor increases to $\gamma = 5$ and $\gamma = 10$, LotBoNC maintains unlabelled-aware accuracies of 72.86 and 60.15, respectively. In contrast, the best competing method drops to 47.54 and 42.91, showing a substantial degradation. For the unlabelled-agnostic metric, LotBoNC reaches 80.05 at $\gamma = 1$, 71.93 at $\gamma = 5$, and 56.07 at $\gamma = 10$, outperforming all baselines across the board. Other methods perform significantly worse, with the closest baseline at 71.62 for $\gamma = 1$, falling to 34.50 at $\gamma = 5$ and 34.60 at $\gamma = 10$. These consistent improvements on unlabelled samples suggest that LotBoNC is particularly well suited to discovering unlabelled botnet classes in imbalanced scenarios.

Class Distribution Prediction. We evaluate the ability of each method to recover the true class distribution under varying levels of imbalance. We conduct experiments under three settings: $\gamma = 1$, $\gamma = 5$, and $\gamma = 10$, where the ground-truth class prior follows an exponential decay with a head-to-tail ratio of γ. When $\gamma = 1$, the true distribution is uniform across all classes. In this case, LotBoNC produces predictions that are closely aligned with the target, while DeepTraffic deviates more noticeably. As the imbalance increases to $\gamma = 5$, DeepTraffic exhibits significant prediction errors, particularly in underestimating several classes. In contrast, LotBoNC remains substantially closer to the true distribution, with all class-level predictions within a reasonable range of the ground-truth values. When $\gamma = 10$, the task becomes more challenging due to the severe skew. DeepTraffic's prediction quality continues to degrade under this setting. LotBoNC also experiences performance drop, but its output remains consistently more faithful to the expected class distribution than the baseline. Across all three imbalance levels, LotBoNC achieves more accurate and reliable recovery of the class prior. The prediction gap between the two methods widens as γ increases, highlighting the advantage of LotBoNC in handling extreme long-tail distributions.

5.6 Impact of Labelled-to-Unlabelled Class Ratio ρ

This experiment evaluates the robustness of our method under different proportions of labelled and unlabelled class categories. Specifically, we vary the labelled-to-unlabelled class ratio, denoted as $\rho = C_L/C_U$, where C_L is the number of classes with available labels and C_U is the number of unlabelled classes during training.

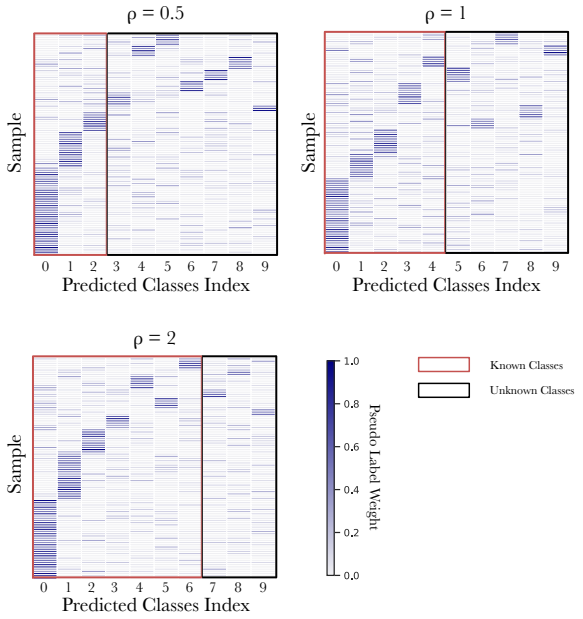


Figure 5: Heatmaps of pseudo-label assignment weights for unlabeled samples under different label-to-unlabeled class ratios $\rho \in \{0.5, 1, 2\}$. Each heatmap shows confidence scores across class indices (columns) for individual samples (rows), with color intensity indicating the pseudo-label weight (darker is higher). Red and black boxes highlight known and unknown class indices, respectively. As ρ increases, dark vertical bands consistently appear within the unknown class region (right side of each heatmap), indicating that our method assigns high-confidence pseudo-labels to unknown tail classes even under severely imbalanced conditions.



Figure 6: t-SNE visualization of IoT-23 embeddings for one head and two tail classes. LotBoNC (right) forms clearer boundaries between head and tail classes, while DeepTraffic (left) shows significant overlap.

A smaller value of ρ reflects a more challenging setup, where the model must generalize from a smaller labelled class set while handling a larger pool of unknown classes. We test three representative ratios: $\rho = 0.5$, $\rho = 1$, and $\rho = 2$. In all settings, we maintain a

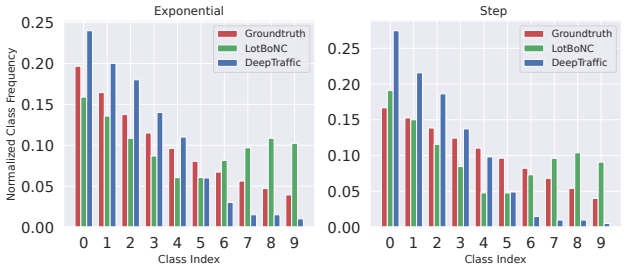


Figure 7: Class-wise pseudo-label distribution recovery comparison for DeepTraffic and LotBoNC against the ground truth under $\gamma \in \{1, 5, 10\}$. LotBoNC reliably tracks long-tailed targets across all imbalance levels, while DeepTraffic collapses minority estimates.

fixed total number of classes and only adjust how many of them are labelled versus unlabelled.

Table 3 summarizes the results across all three datasets. When $\rho = 0.5$, meaning only one labelled class for every two unlabelled classes, the method already achieves strong performance. For example, the all-class accuracy reaches 83.56 on *IoT-23*, 78.37 on *ISCX-2014*, and 81.55 on *CTU-13*. Despite the limited supervision, the accuracy on the labelled subset remains consistently above 98.5 across all datasets, indicating that our method preserves head-class performance even under reduced label coverage. When the ratio is increased to $\rho = 1$, where the number of labelled and unlabelled classes is equal, we observe a consistent improvement across all evaluation settings. The all-class accuracy improves to 88.88 on *IoT-23*, 84.22 on *ISCX-2014*, and 87.89 on *CTU-13*. More importantly, the accuracy on the unlabelled-aware setting shows substantial gains, reaching 84.01 on IoT-23 and 82.46 on *CTU-13*. This indicates that including more labelled categories helps the model construct a more structured latent space that benefits generalization to unseen classes. Further increasing the ratio to $\rho = 2$ continues to enhance performance, though with diminishing returns. The all-class accuracy reaches 94.41 on *IoT-23* and 93.60 on *CTU-13*, while unlabelled-aware accuracy improves to 92.59 and 91.33, respectively. The performance gap between unlabelled-aware and unlabelled-agnostic settings narrows as the ratio increases, suggesting that the model becomes increasingly capable of uncovering the structure of unseen classes even when they are not explicitly labelled. To better understand this behavior, Figure 5 visualizes the pseudo-label assignment weights for unlabeled samples under each ρ setting. Each heatmap displays the predicted confidence scores across all class indices, where each row represents a sample and each column corresponds to a predicted class. Higher intensity (darker) values indicate stronger assignment confidence. The known and unknown class indices are marked with red and black vertical borders, respectively. As ρ increases, the model assigns stronger and more structured pseudo-label weights to unknown classes, reflecting its improved ability to generalize to novel categories. This visualization complements the quantitative results by revealing how the latent space becomes increasingly disentangled and class-separable as more labelled classes are introduced.

ρ	Dataset	Overall	Labelled	Unlabelled-Aw†	Unlabelled-Ag†
0.5	IoT-23	83.56 ± 0.76	99.23 ± 0.08	75.73 ± 0.20	73.54 ± 0.60
	ISCX-2014	78.37 ± 0.02	98.57 ± 0.64	68.27 ± 0.79	67.79 ± 0.61
	CTU-13	81.55 ± 0.49	98.93 ± 0.38	72.86 ± 0.57	71.93 ± 0.55
1	IoT-23	88.88 ± 0.19	98.61 ± 0.41	84.01 ± 0.78	82.82 ± 0.13
	ISCX-2014	84.22 ± 0.57	97.39 ± 0.71	77.64 ± 0.68	76.70 ± 0.33
	CTU-13	87.89 ± 0.70	98.75 ± 0.08	82.46 ± 0.59	80.96 ± 0.06
2	IoT-23	94.41 ± 0.69	98.04 ± 0.48	92.59 ± 0.47	85.34 ± 0.31
	ISCX-2014	88.79 ± 0.72	97.74 ± 0.46	84.31 ± 0.69	77.68 ± 0.11
	CTU-13	93.60 ± 0.52	98.14 ± 0.16	91.33 ± 0.49	85.30 ± 0.51

Table 3: Performance of LotBoNC under different labelled-to-unlabelled class ratios. † "Aw" stands for Aware and "Ag" stands for Agnostic.

5.7 Ablation Study

5.7.1 Impact of class imbalance distribution. To assess the impact of unknown-class distribution shape, we perform an ablation study where the exponential long-tailed distribution is replaced with a step-decreasing alternative, following the protocol introduced in LDAM [6]. This variant groups unknown classes into buckets with equal sample counts, decreasing in discrete steps. Importantly, we keep the total number of unknown samples fixed via the same $\gamma = n_k/n_u$, ensuring that the global class ratio remains unchanged. This isolates the effect of distribution shape alone. Table 4 reports the overall and unlabeled accuracy across three datasets and three imbalance levels ($\gamma \in \{1, 5, 10\}$), comparing our method and a representative baseline (DeepTraffic). We observe that switching to a step-decreasing histogram consistently improves performance across all settings. The gains are especially pronounced on the unlabelled metric, with improvements ranging from 3–5% for both DeepTraffic and LotBoNC. This demonstrates that models benefit from reducing the severity of the extreme tail classes. In addition, the relative performance gap remains consistent that LotBoNC continues to outperform the strongest baseline under both distribution shapes and all imbalance settings. These findings suggest that while distribution shape affects absolute accuracy, LotBoNC's advantage is robust across different structures.

5.7.2 Impact of Pre-training. We assess the effectiveness of pre-training in improving novel class discovery under varying levels of imbalance ($\gamma \in \{1, 5, 10\}$). As summarized in Table 5, pre-training consistently enhances performance across all datasets and imbalance settings. For instance, at $\gamma = 1$, the unlabelled-agnostic accuracy improves from 60.21% to 75.73% on *IoT-23*, from 52.18% to 68.27% on *ISCX-2014*, and from 58.34% to 72.86% on *CTU-13*. The improvements are even more substantial at higher imbalance levels; at $\gamma = 10$, accuracy increases from 77.82% to 92.59% on *IoT-23*, from 66.48% to 84.31% on *ISCX-2014*, and from 75.10% to 91.33% on *CTU-13*. These results confirm that our pre-training is crucial for discovering novel botnet under severe long-tailed conditions.

5.7.3 Impact of Loss Ratio. We study the influence of the loss ratio λ_{proto}, which controls the trade-off between prototype-level supervision and instance-level contrastive learning. We vary $\lambda \in \{0.1, 0.3, 0.5, 0.7, 1.0\}$ across three imbalance scenarios defined by $\gamma \in \{1, 5, 10\}$, where a higher γ indicates a more severe long-tailed distribution in the unlabeled data. The results are reported in Figure 8 over four evaluation metrics: All, Labelled, Unlabelled-Aware,

γ	Metric	IoT-23				ISCX-2014				CTU-13			
		DeepTraffic		LotBoNC		DeepTraffic		LotBoNC		DeepTraffic		LotBoNC	
		Exponential	Step	Exponential	Step	Exponential	Step	Exponential	Step	Exponential	Step	Exponential	Step
1	Overall	59.22 ± 0.61	61.22 ± 0.60	89.88 ± 0.19	90.88 ± 0.19	53.53 ± 0.07	55.53 ± 0.07	85.72 ± 0.57	86.52 ± 0.57	60.78 ± 0.64	62.78 ± 0.64	88.89 ± 0.70	91.39 ± 0.70
	Unlabelled	31.01 ± 0.10	35.00 ± 0.55	63.01 ± 0.78	66.01 ± 0.80	25.21 ± 0.43	28.50 ± 0.40	58.23 ± 0.68	62.23 ± 0.70	32.60 ± 0.38	36.00 ± 0.38	61.84 ± 0.59	66.84 ± 0.60
5	Overall	63.50 ± 0.61	65.00 ± 0.61	91.30 ± 0.19	92.30 ± 0.19	56.80 ± 0.07	58.20 ± 0.07	88.30 ± 0.57	89.30 ± 0.57	63.90 ± 0.64	65.30 ± 0.64	90.90 ± 0.70	92.00 ± 0.70
	Unlabelled	20.15 ± 0.10	23.50 ± 0.10	40.96 ± 0.78	44.50 ± 0.80	17.65 ± 0.43	20.50 ± 0.43	40.76 ± 0.68	43.00 ± 0.70	22.82 ± 0.38	25.80 ± 0.38	43.29 ± 0.59	46.00 ± 0.60
10	Overall	68.00 ± 0.61	69.20 ± 0.61	92.90 ± 0.19	93.90 ± 0.19	60.40 ± 0.07	61.60 ± 0.07	90.20 ± 0.57	91.20 ± 0.57	67.30 ± 0.64	68.50 ± 0.64	92.90 ± 0.70	94.00 ± 0.70
	Unlabelled	13.00 ± 0.10	16.00 ± 0.10	29.00 ± 0.78	32.00 ± 0.80	12.50 ± 0.43	15.00 ± 0.43	30.50 ± 0.68	33.00 ± 0.70	14.10 ± 0.38	17.00 ± 0.38	30.10 ± 0.59	33.50 ± 0.60

Table 4: For each imbalance level γ, we report performance on overall accuracy and on the unlabeled accuracy. Each dataset compares baseline Deeptraffic with our LotBoNC under the exponential and step-decreasing distribution.

γ	Setting	IoT-23	ISCX-2014	CTU-13
1	w/ Pre-train	88.20	82.55	80.05
	w/o Pre-train	74.21	72.18	68.34
5	w/ Pre-train	73.54	67.79	71.93
	w/o Pre-train	65.75	58.13	60.89
10	w/ Pre-train	56.05	52.78	56.07
	w/o Pre-train	47.82	41.23	45.19

Table 5: Unlabelled-agnostic accuracy (%) of evaluating the impact of the pre-training across varying imbalance ratios ($\gamma \in 1, 5, 10$). Pre-training yields substantial improvements in novel class discovery when class imbalance increases.

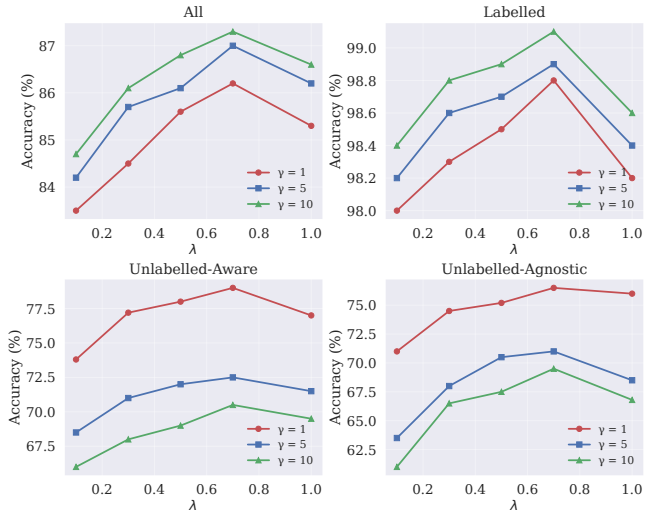


Figure 8: Impact of prototype loss ratio λ on performance under different label-unlabelled class ratios ρ. Higher λ consistently improves unknown-aware and unknown-agnostic accuracy, indicating stronger prototype supervision enhances novel class discovery.

and Unlabelled-Agnostic. We observe that as the class imbalance increases from $\gamma = 1$ to $\gamma = 10$, the unlabelled-aware and unlabelled-agnostic accuracies gradually decrease, reflecting the increasing difficulty in discovering novel tail classes. Conversely, the labelled and overall accuracies show a slight increase, as the higher imbalance implies a larger volume of labeled data relative to unknowns. This trend is consistent across all loss ratios. Across all three imbalance levels, we find that $\lambda_{\text{proto}} = 0.7$ yields the best performance in every metric. For instance, under the balanced setting ($\gamma = 1$), increasing λ from 0.1 to 0.7 improves the unlabelled-aware accuracy from 73.8% to 79.5%, and unlabelled-agnostic accuracy from 71.0% to 76.5%, while maintaining high labelled accuracy (above 98.0%) and improving the all-class accuracy from 83.5% to 86.2%. Similar improvements are observed at $\gamma = 5$, where unlabelled-aware accuracy rises from 66.5% to 74.0%, and at $\gamma = 10$, from 64.0% to 72.0%, with labelled accuracy exceeding 98.9% in both cases. Moreover, the difference in overall accuracy across λ remains within three percentage points for each γ, suggesting that the model is stable to moderate changes in this hyperparameter. Nevertheless, the consistent peak at $\lambda_{\text{proto}} = 0.7$ highlights the importance of moderately strong prototype supervision in aligning pseudo-labels with evolving class prototypes, especially in imbalanced open-world settings.

5.7.4 Impact of Loss Components. We explore the contribution of each loss component in our framework under varying class imbalance settings defined by $\gamma \in \{1, 5, 10\}$. Specifically, we remove one loss term at a time: the prototype-based loss ($\mathcal{L}_{\text{proto}}$), the instance-level contrastive loss ($\mathcal{L}_{\text{ins}}$), and the supervised loss ($\mathcal{L}_{\text{sup}}$), and compare each variant to the full LotBoNC model. Results across all four evaluation metrics are reported in Figure 9. Across all imbalance levels, the full model outperforms its ablated variants. At $\gamma = 1$, where class distribution is balanced, removing the supervised loss leads to the most severe degradation: all-class accuracy drops from 86.2% to 61.0%, and unknown-aware accuracy falls from 79.5% to 46.0%. This shows that supervision from labeled samples is crucial to anchor representation learning even in balanced scenarios. Removing $\mathcal{L}_{\text{proto}}$ or $\mathcal{L}_{\text{ins}}$ also weakens performance, reducing unlabelled-aware accuracy to 73.0% and 50.0%, respectively. As the imbalance intensifies at $\gamma = 5$, the gap becomes more pronounced. LotBoNC achieves 74.0% unknown-aware accuracy, while its no-$\mathcal{L}_{\text{sup}}$ variant drops to 28.0%, and no-$\mathcal{L}_{\text{ins}}$ variant to 35.0%. The full model also maintains a high labelled accuracy of 99.1%, compared to 75.0% without $\mathcal{L}_{\text{sup}}$. This underscores the importance of both

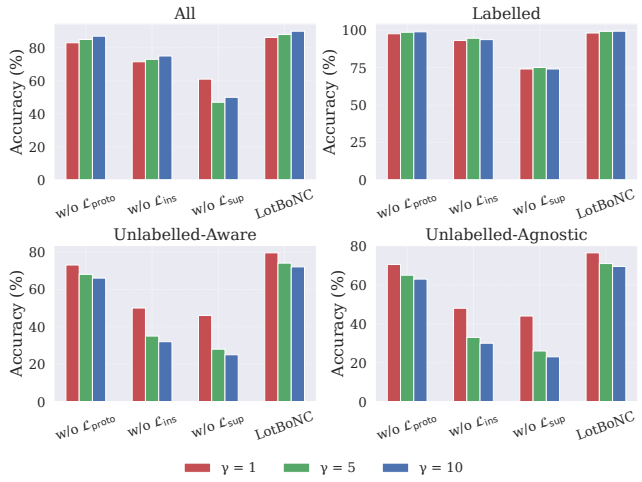


Figure 9: Ablation of loss components under varying label-unlabelled class ratios $\gamma \in \{1, 5, 10\}$. Removing any component degrades performance, with the largest drop observed when omitting supervised loss ($\mathcal{L}_{\text{sup}}$). Prototype loss ($\mathcal{L}_{\text{proto}}$) is essential for novel class separation, while instance contrastive loss ($\mathcal{L}_{\text{ins}}$) aids stability under heavy imbalance.

supervised grounding and instance-level consistency when facing long-tailed unknown-class distributions. At $\gamma = 10$, the hardest setting, we observe similar patterns. The full method maintains 72.0% unknown-aware accuracy and 69.5% unknown-agnostic accuracy. In contrast, omitting $\mathcal{L}_{\text{sup}}$ leads to a substantial drop to 25.0% and 23.0%, respectively. Meanwhile, removing $\mathcal{L}_{\text{ins}}$ still causes a notable degradation to 32.0% in unknown-aware accuracy. Overall, while all three components contribute meaningfully especially in severely imbalanced settings. These results validate the necessity of combining global prototype alignment, local contrastive regularization, and supervised signals to robustly discover novel classes under open-world imbalance.

5.7.5 Impact of Encoder Dropout Rate. We evaluate the effect of encoder-level stochasticity by varying the dropout rate across four values: 0, 0.05, 0.10, and 0.50. while keeping all other settings fixed. Experiments are conducted on three datasets under imbalance levels of 1,5 and10, consistent with the previous experiments. As shown in Table 6, all datasets exhibit a consistent trend: performance is lowest at $p = 0$, improves with mild dropout, and peaks at $p = 0.10$. Heavier noise ($p = 0.50$) leads to a clear decline. These results confirm that moderate dropout is essential for learning stable representations, while excessive noise impairs generalization particularly under class imbalance. We therefore adopt $p = 0.10$ as the default in all experiments.

6 Ethical Considerations and Open Source

All experiments in this study were conducted exclusively on publicly available datasets were released by academic and research institutions under licenses permitting non-commercial research. We strictly adhere to their respective usage terms and confirm that no private, personally identifiable, or proprietary information is involved. No traffic was intercepted, collected, or analyzed outside the scope of these public datasets. LotBoNC is intended to improve the robustness and reliability of network defense systems through automated discovery of novel botnet behaviors. However, we acknowledge that any traffic classification system, if misused, may pose ethical risks such as non-consensual monitoring, profiling, or censorship. To mitigate this, we emphasize that our framework is designed for deployment in regulated, legally compliant environments such as enterprise security operations centers, academic research networks, or CERTs, where oversight and data governance structures are in place. We explicitly discourage the application of this research in surveillance infrastructures that lack transparent accountability mechanisms or operate outside democratic oversight. While our system does not target user content and operates solely on encrypted flow-level metadata, we recognize that any form of automated traffic analysis must be handled with care to avoid unintended consequences.

To support responsible innovation and reproducibility, we will release related code and scripts upon publication. We encourage the research community to build upon our work within the bounds of ethical, lawful, and privacy-respecting usage.

p	IoT-23			ISCX			CTU-13		
	γ=1	5	10	1	5	10	1	5	10
0.00	58.2	61.3	63.5	60.2	62.5	64.8	62.0	64.5	66.8
0.05	88.1	89.7	90.9	83.9	86.0	88.0	87.0	89.2	91.4
0.10	**89.9**	**91.3**	**92.9**	**85.7**	**88.3**	**90.2**	**88.9**	**90.9**	**92.9**
0.50	70.1	71.5	72.3	65.6	67.3	69.0	69.0	70.7	72.4

Table 6: Overall accuracy for different dropout rates p, and imbalance levels γ.

7 Conclusion

This work introduces LotBoNC, a framework for long-tailed novel botnet traffic classification under open-world conditions. By integrating entropy-regularized optimal transport with prototype refinement, LotBoNC achieves calibrated pseudo-label assignments that preserve tail semantics while avoiding collapse into dominant known classes. Our extensive evaluations across diverse traffic domains demonstrate consistent state-of-the-art performance, especially under severe imbalance and low supervision. Beyond classification accuracy, LotBoNC exhibits robustness in discovering novel botnet clusters, setting a new baseline for practical open-world encrypted traffic analysis.

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A Appendix

A.1 Prototype Loss for Unlabeled Data

We aim to learn the model parameters θ that maximize the log-likelihood over a set of m unlabeled samples:

$$\theta^* = \arg\max_{\theta} \sum_{i=1}^{m} \log p\left(x^{(i)} \mid \theta\right) \tag{15}$$

Here, the observed data $\left\{x^{(i)}\right\}_{i=1}^{m}$ is assumed to be associated with latent variables $C = \{c_k\}_{k=1}^{|\mathcal{Y}_u|}$, where each c_k denotes a prototype representing a class. The total set of classes in the unlabeled data is denoted by $\mathcal{Y}_u$, which includes both the known classes $\mathcal{Y}_l$ from the labeled data and the additional unknown classes $\mathcal{Y}_n = \mathcal{Y}_u \backslash \mathcal{Y}_l$.

The joint distribution of each input x_i with its associated prototype c_i can be expressed as:

$$\begin{aligned}\theta^* &= \arg\max_{\theta} \sum_{i=1}^{m} \log p\left(x^{(i)} \mid \theta\right) \\ &= \arg\max_{\theta} \sum_{i=1}^{m} \log \sum_{c_k \in C} p\left(x^{(i)}, c_k^{(i)} \mid \theta\right)\end{aligned} \tag{16}$$

Directly optimizing this objective is intractable, so we instead optimize a lower bound via variational approximation:

$$\begin{aligned}&\sum_{i=1}^{m} \log \sum_{c_k \in C} p\left(x^{(i)}, c_k^{(i)} \mid \theta\right) \\ &= \sum_{i=1}^{m} \log \sum_{c_k \in C} Q\left(c_k^{(i)}\right) \frac{p\left(x^{(i)}, c_k^{(i)} \mid \theta\right)}{Q\left(c_k^{(i)}\right)} \\ &\geq \sum_{i=1}^{m} \sum_{c_k \in C} Q\left(c_k^{(i)}\right) \log \frac{p\left(x^{(i)}, c_k^{(i)} \mid \theta\right)}{Q\left(c_k^{(i)}\right)}\end{aligned} \tag{17}$$

In the above, $Q\left(c_k^{(i)}\right)$ represents a variational distribution over the prototypes. The inequality follows from Jensen's inequality [22]. Equality holds when the ratio $\frac{p\left(x^{(i)}, c_k^{(i)} \mid \theta\right)}{Q\left(c_k^{(i)}\right)}$ is constant, which leads to:

$$\begin{aligned}Q\left(c_k^{(i)}\right) &= \frac{p\left(x^{(i)}, c_k^{(i)} \mid \theta\right)}{\sum_{c_k \in C} p\left(x^{(i)}, c_k^{(i)} \mid \theta\right)} \\ &= \frac{p\left(x^{(i)}, c_k^{(i)} \mid \theta\right)}{p\left(x^{(i)} \mid \theta\right)} \\ &= p\left(c_k^{(i)} \mid x^{(i)}, \theta\right)\end{aligned} \tag{18}$$

By omitting the constant entropy term $-\sum_{i=1}^{n} \sum_{c_k \in C} Q\left(c_k^{(i)}\right) \log Q\left(c_k^{(i)}\right)$, the surrogate objective becomes:

$$\sum_{i=1}^{m} \sum_{c_k \in C} Q\left(c_k^{(i)}\right) \log p\left(x^{(i)}, c_k^{(i)} \mid \theta\right) \tag{19}$$

Using the E-step result, we then optimize Eq. 19 with respect to θ:

$$\begin{aligned}&\sum_{i=1}^{n} \sum_{c_k \in C} Q\left(c_k^{(i)}\right) \log p\left(x^{(i)}, c_k^{(i)} \mid \theta\right) \\ &= \sum_{i=1}^{n} \sum_{c_k \in C} p\left(c_k^{(i)} \mid x^{(i)}, \theta\right) \log p\left(x^{(i)}, c_k^{(i)} \mid \theta\right) \\ &= \sum_{i=1}^{n} \sum_{c_k \in C} \underbrace{\mathbb{1}\left(x^{(i)} \in c_k^{(i)}\right)}_{k^* = \arg\max_k A[i][k], \text{then } x^{(i)} \in c_{k^*}} \log p\left(x^{(i)}, c_k^{(i)} \mid \theta\right)\end{aligned} \tag{20}$$

Furthermore, we decompose the joint probability as:

$$p\left(x^{(i)}, c_k^{(i)} \mid \theta\right) = p\left(x^{(i)} \mid c_k^{(i)}, \theta\right) p\left(c_k^{(i)} \mid \theta\right)$$

Following the formulation in [31], we model $p\left(x^{(i)} \mid c_k^{(i)}, \theta\right)$ as an isotropic Gaussian centered at the prototype $c_k^{(i)}$, with shared variance σ:

$$\begin{aligned}&p\left(x^{(i)} \mid c_k^{(i)}, \theta\right) \\ &= \exp\left(\frac{-\left(v_i - c_k^{(i)}\right)^2}{2\sigma^2}\right) / \sum_{j=1}^{|C|} \exp\left(\frac{-\left(v_i - c_j^{(i)}\right)^2}{2\sigma^2}\right),\end{aligned} \tag{21}$$

Here, $v_i = f_\theta(x^{(i)})$ denotes the embedding of input $x^{(i)}$, and we apply ℓ_2 normalization to both v_i and each prototype $c_k^{(i)}$. Substituting into the likelihood, the optimization problem becomes:

$$\begin{aligned}\theta^* = \arg\min_{\theta} \sum_{i=1}^{n} (&- \log \frac{\exp\left(v_i \cdot c_k^{(i)}\right)}{\sum_{j=1}^{|\mathcal{Y}_u|} \exp\left(v_i \cdot c_j^{(i)}\right)} \\ &- \log p\left(c_k^{(i)} \mid \theta\right))\end{aligned} \tag{22}$$

This leads to the following prototype-based loss for unlabeled instances:

$$\mathcal{L}_{proto}^{(i)} = -\log \frac{\exp\left(v_i \cdot c_k^{(i)}\right)}{\sum_{j=1}^{k} \exp\left(v_i \cdot c_j^{(i)}\right)} - \log p\left(c_k^{(i)} \mid \theta\right) \tag{23}$$